Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/12/2006
PI Name:	Blawzdziewicz, Jerzy Ph.D.		
Project Title:	Drop Coalescence: Drainage and Stabilization of Thin Liquid Films		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8286	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/05/2002	**End Date:**	11/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rame, Enrique (National Center for Microgravity Research)		
Grant/Contract No.:	NAG3-2704		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Theory: we will develop asymptotic theories and computer simulation methods to describe the drainage and stabilization of thin liquid films. Experiments: we will demonstrate the stabilization mechanism due to external flows. To develop theories of long-time drainage of liquid films with tangentially-mobile interfaces. To investigate theoretically and analytically the effect of external flows on film drainage. To examine the dynamics of film rupture when van der Waals forces are important. To describe dynamical effects of surfactants adsorbed on film interfaces and colloidal particles or micelles present in the film. Elucidate new non-coalescence phenomenon and resolve controversies about the effect of ambient flows on film drainage dynamics. Identify factors that control drop stability in emulsion flows. Evaluate coalescence rates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Dynamics of thin liquid films: Our results can be used in developing improved film stabilization methods. Moreover our study may lead to new techniques for measuring transverse structural forces in liquid films. Suspension flow in thin films and slit pores: Our method can be used in computer simulations of the dynamical behavior of confined suspensions, in studies of motion of polymer molecules in narrow channels, and in simulations of collective motions in bacterial colonies. Other potential applications include development of new methods for particle separation, techniques for controlled adsorption of particle arrays on patterned walls in microfluidic devices, as well as predictions of suspension flows in porous materials (e.g., propagation of contaminants in soil).
Task Progress:	1. Dynamics of thin liquid films a) Using thin-film approach, boundary-integral simulations, and asymptotic analysis we have shown that drops can be stabilized to coalescence by external flow. Intermediate- and long-time behavior of the thin film separating the drops have been described, and the role of van der Waals forces has been examined. Our results can be used to predict drop coalescence rates in emulsion flows. They will also help to develop better methods of drop manipulation in microfluidic devices. b) We have developed a quantitative theory for description of the stepwise drainage of thin liquid films stabilized by colloidal particles. We have demonstrated that thin film regions of different thickness can coexist in full thermodynamical equilibrium. A new method for predicting the phase-equilibrium conditions have been proposed. Our results can be used in developing improved film stabilization methods. Moreover our study may lead to new techniques for measuring transverse structural forces in liquid films. 2. Dynamics of liquid drops We have provided numerical and theoretical descriptions of drop deformation in external flows. We have also developed a phenomenological theory for drop evolution and breakup in flows of near-critical strengths. In particular, we have predicted and explained a universal slow drop evolution under near-critical conditions. The dependence of the the sizes of drop fragments on flow strength have been described. Our results can be used in analyses of drop breakup during emulsification process and in development of new polymer-blend-based microstructured materials. 3. Suspension flow in thin films and slit pores We have developed a new, highly accurate algorithm for evaluation of particle motion in suspensions confined in thin liquid films and slit pores. Our method can be used in computer simulations of the dynamical behavior of confined suspensions, in studies of motion of polymer molecules in narrow channels, and in simulations of collective motions in bacterial colonies. Other potential applications include development of new methods for particle separation, techniques for controlled adsorption of particle arrays on patterned walls in microfluidic devices, as well as predictions of suspension flows in porous materials (e.g., propagation of contaminants in soil). The effect of confinement on the rheology of a dilute suspension of non-spherical particles has also been described. The PI is also coordinating the Fluid-Dynamics sessions for the 2006 AIChE Annual Meeting.
Bibliography Type:	Description: (Last Updated: 04/23/2010)
Articles in Peer-reviewed Journals	Bhattacharya S, Blawzdziewicz J, Wajnryb E. "Far-field approximation for hydrodynamic interactions in parallel-wall geometry." Journal of Computational Physics. 2006 Mar 1;212(2):718-38. , Mar-2006
Articles in Peer-reviewed Journals	Bhattacharya S, Blawzdziewicz J, Wajnryb E. "Hydrodynamic interactions of spherical particles in suspensions confined between two planar walls." Journal of Fluid Mechanics. 2005 Oct;541:263-92. , Oct-2005
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Articles in Peer-reviewed Journals	Blawzdziewicz J, Wajnryb E. "Phase equilibria in stratified thin liquid films stabilized by colloidal particles." Europhys Lett. 2005 July;71(2):269-75. , Jul-2005
Articles in Peer-reviewed Journals	Blawzdziewicz J, Wajnryb E, Given J, Hubbard JB. "Sharp scalar and tensor bounds on the hydrodynamic friction and mobility of arbitrarily shaped bodies in Stokes flow." Physics of Fluids. 2005 Mar;17:033602. , Mar-2005
Articles in Peer-reviewed Journals	Bhattacharya S, Blawzdziewicz J, Wajnryb E. "Hydrodynamic interactions of spherical particles in Poiseuille flow between two parallel walls." Physics of Fluids, in press, 2006. , Jul-2006
Books/Book Chapters	Blawzdziewic J. "Boundary integral methods for Stokes Flows." in "Computational Methods for Multiphase Flow." Ed. A. Prosperetti and G. Tryggvason. Cambridge University Press, In Press, 2006 , Jul-2006